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College of Engineering
& Computer Science

**Multiple Autonomous
Robot Systems (MARS) Lab**



INTRODUCTION OF A NEW PATH-FINDING BUG ALGORITHM: COMBOBUG

Izabella Valero and Qi Lu

Department of Computer Science

The University of Texas Rio Grande
Valley

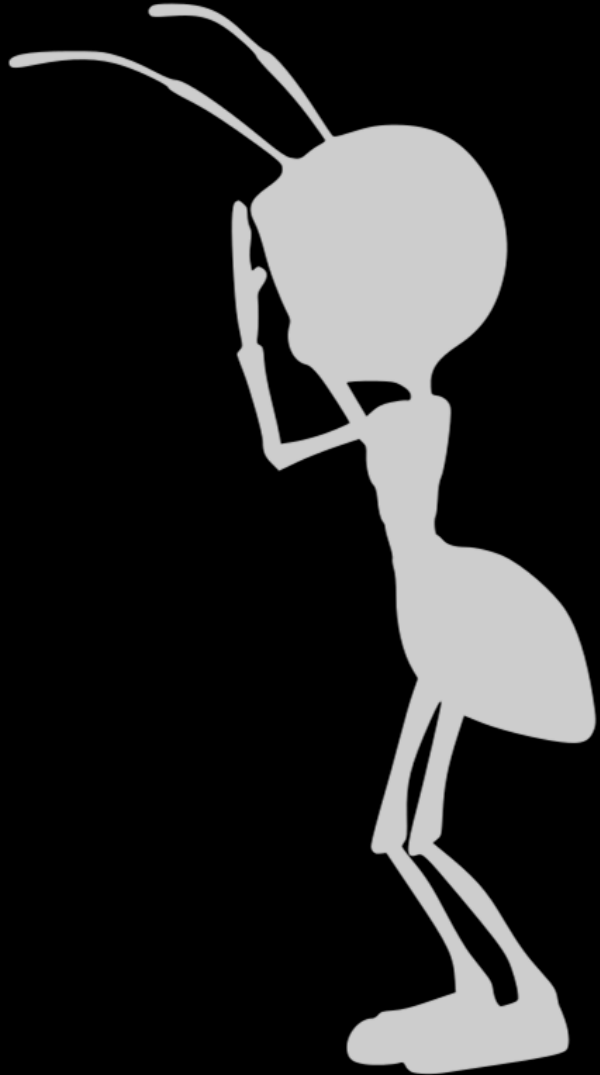


WHAT IS A BUG ALGORITHM?

Bug Algorithms are obstacle avoidance algorithms for navigating an environment and reaching a goal.

- Deterministic Behavior for Minimal Computation Time
- Localization for Unknown Dynamic Environments





CLOSE YOUR EYES

Let's imagine for a minute.

WHY DO WE NEED THEM?

Search and Rescue

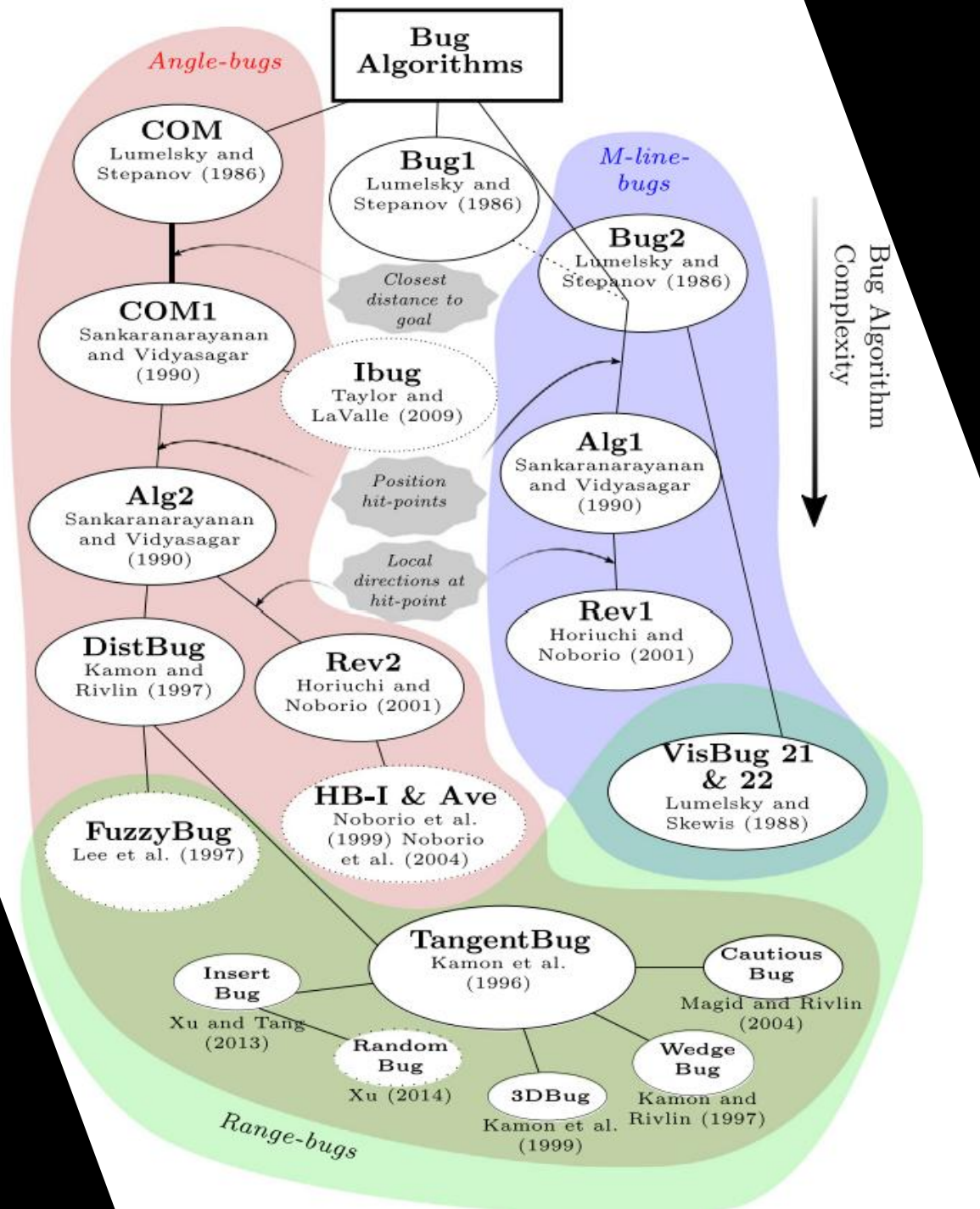
Disaster Zones

Warehouse robots

Home Improvement Robots

Expansion of Obstacle Avoidance Algorithms





BUG DISTINCTIONS

Each of these strategies can be elaborated on and have a set of algorithms that utilize them.

- Common-Sense
- M-Line



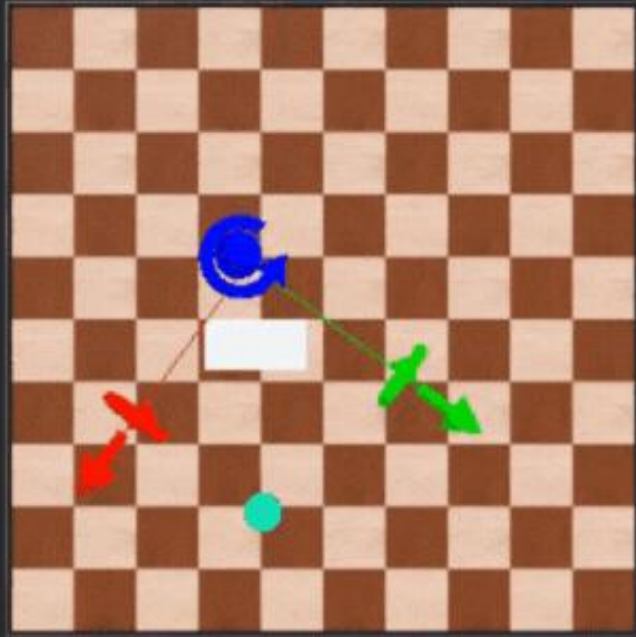
COMMON SENSE

- Move towards goal when there is no obstacle between robot and goal
- Obstacle found? Wall-Follow!
- Greedy Path
- Includes Bug0 (aka Com), Alg2, and Rev2 algorithms



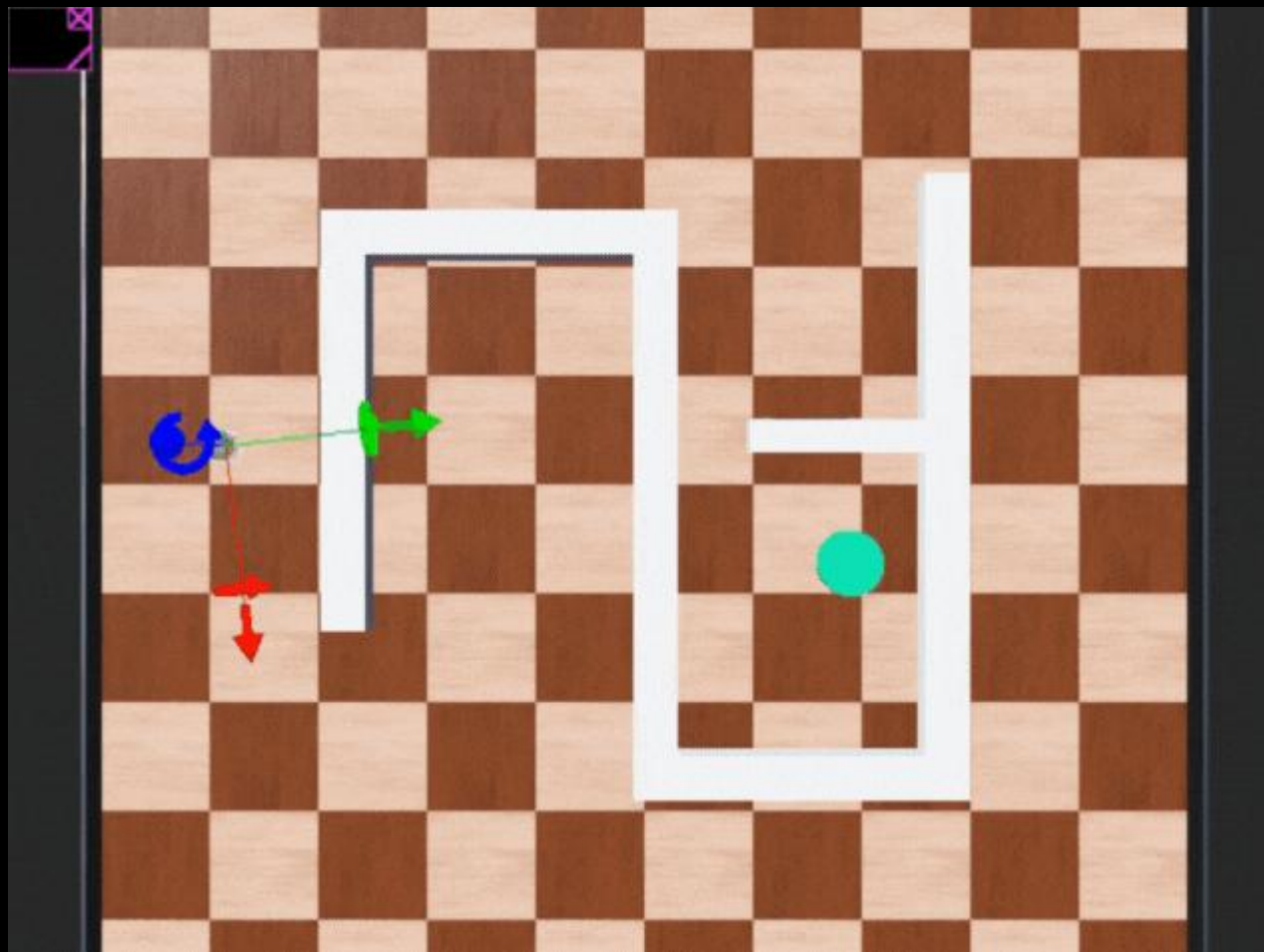
M-LINE

- Calculate M-Line and follow predetermined path
- Obstacle found? Wall-Follow until M-line is found again!
- Includes Bug2, Alg1, and Rev1 algorithms

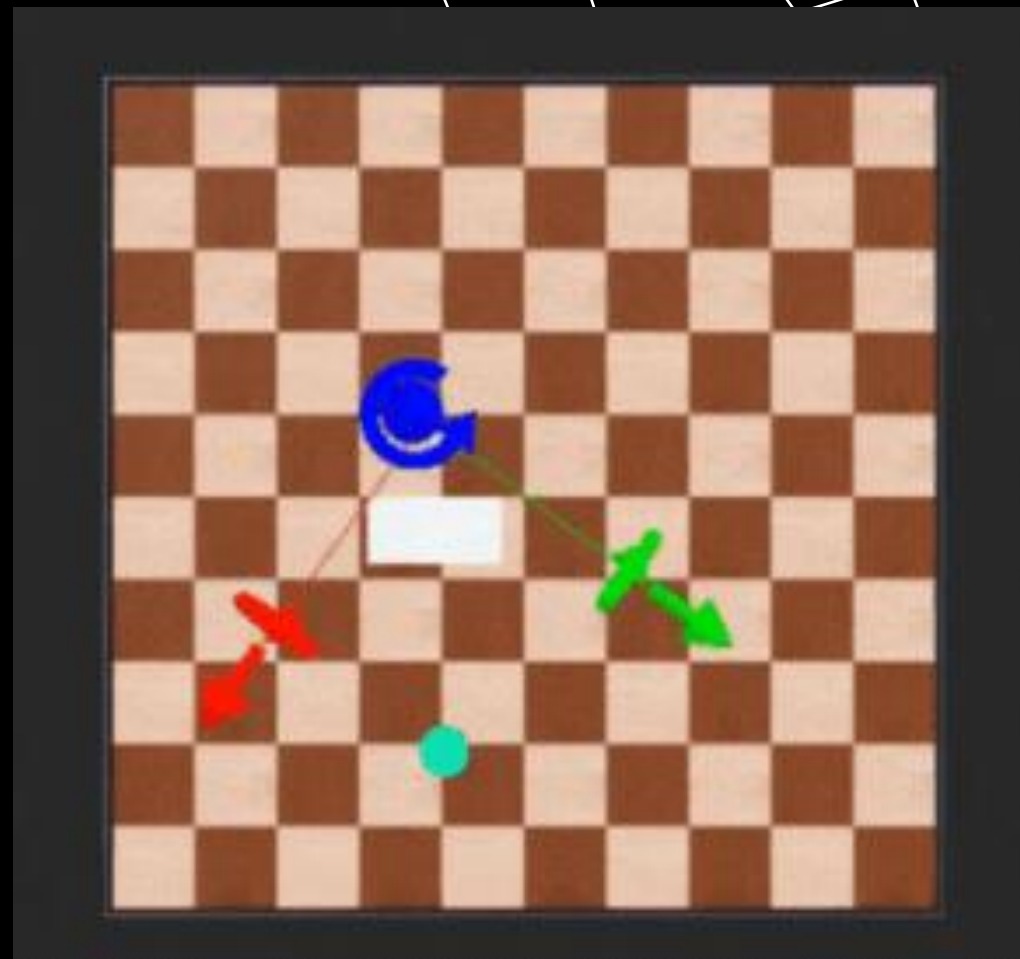


An effective and an ineffective display of Bug0

BUG ALGORITHM AT WORK



An effective and ineffective path of Bug2



BUG ALGORITHM AT WORK

THE PROBLEM

They are effective algorithms, but we can always try to improve upon the inefficiencies that occur to obtain a better path to the goal.

What do we want to improve?

- Existing algorithms
- Excessive paths
- Redundant paths

What do we want to keep?

- Low computational ability
- Localization





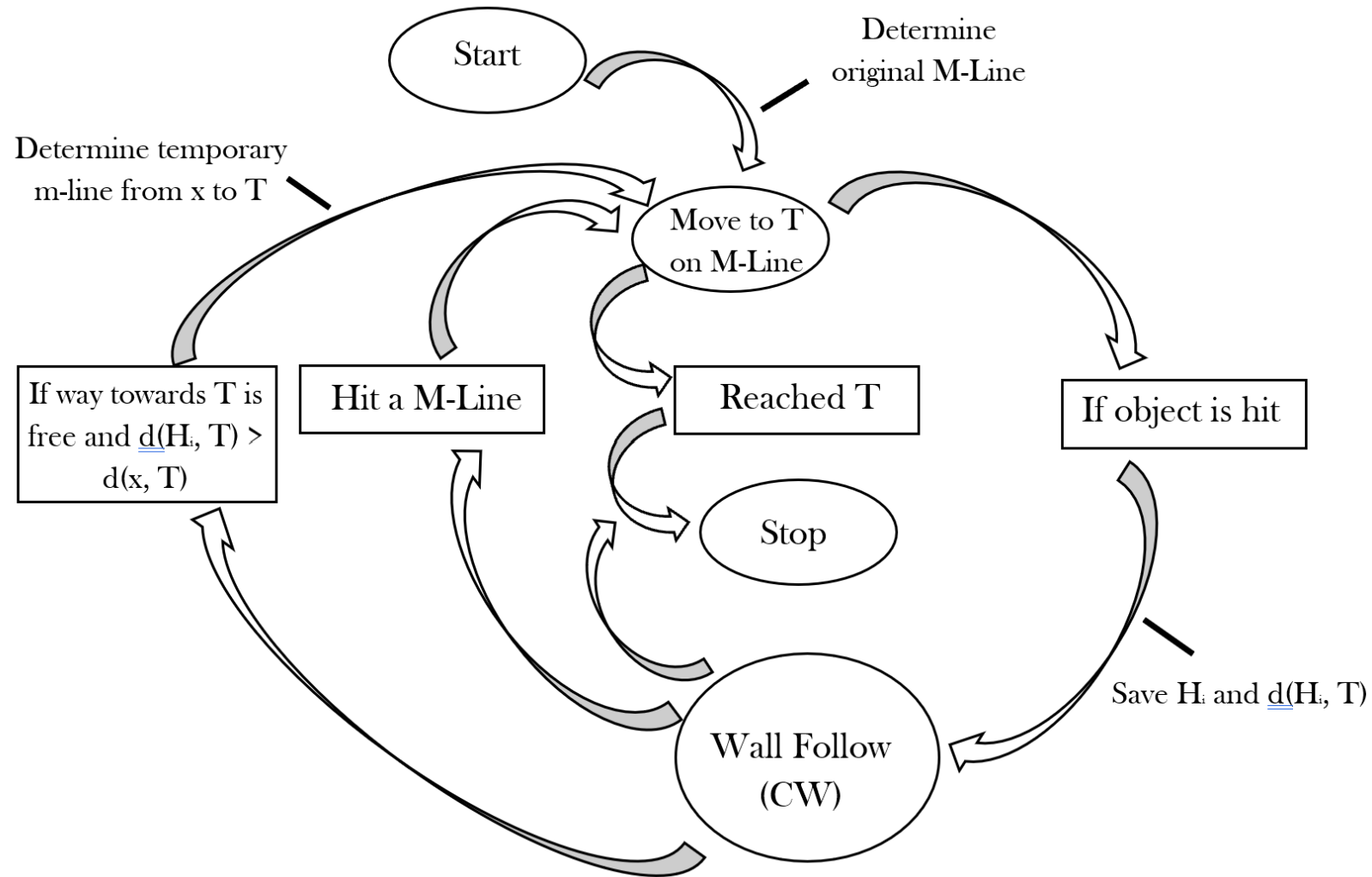
PROPOSED IDEA?

Create new paths/solutions to solve inefficient paths

How?

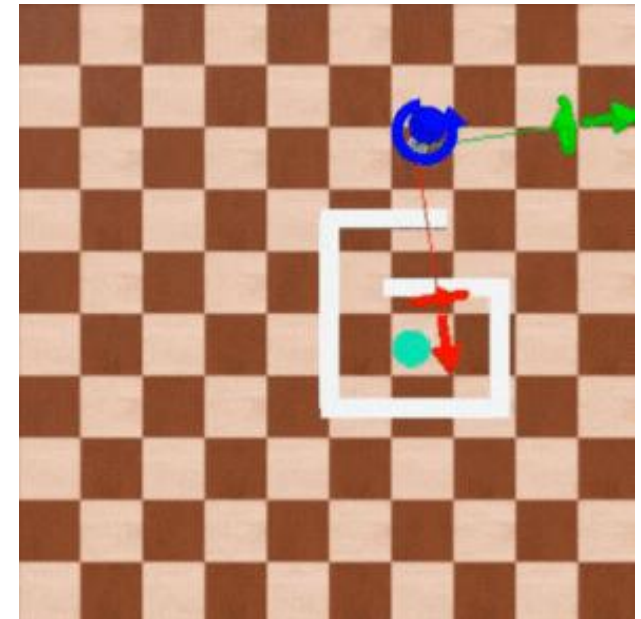
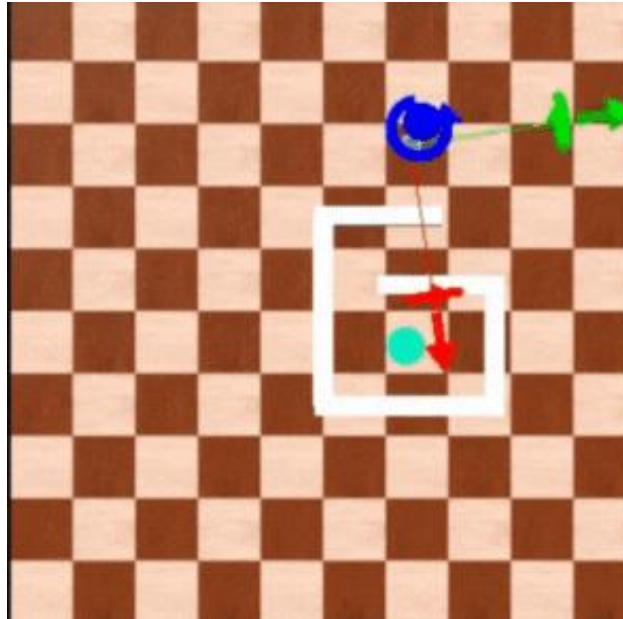
Combining Common-Sense and M-Line strategies to correct where failures occur

COMBOBUG METHOD



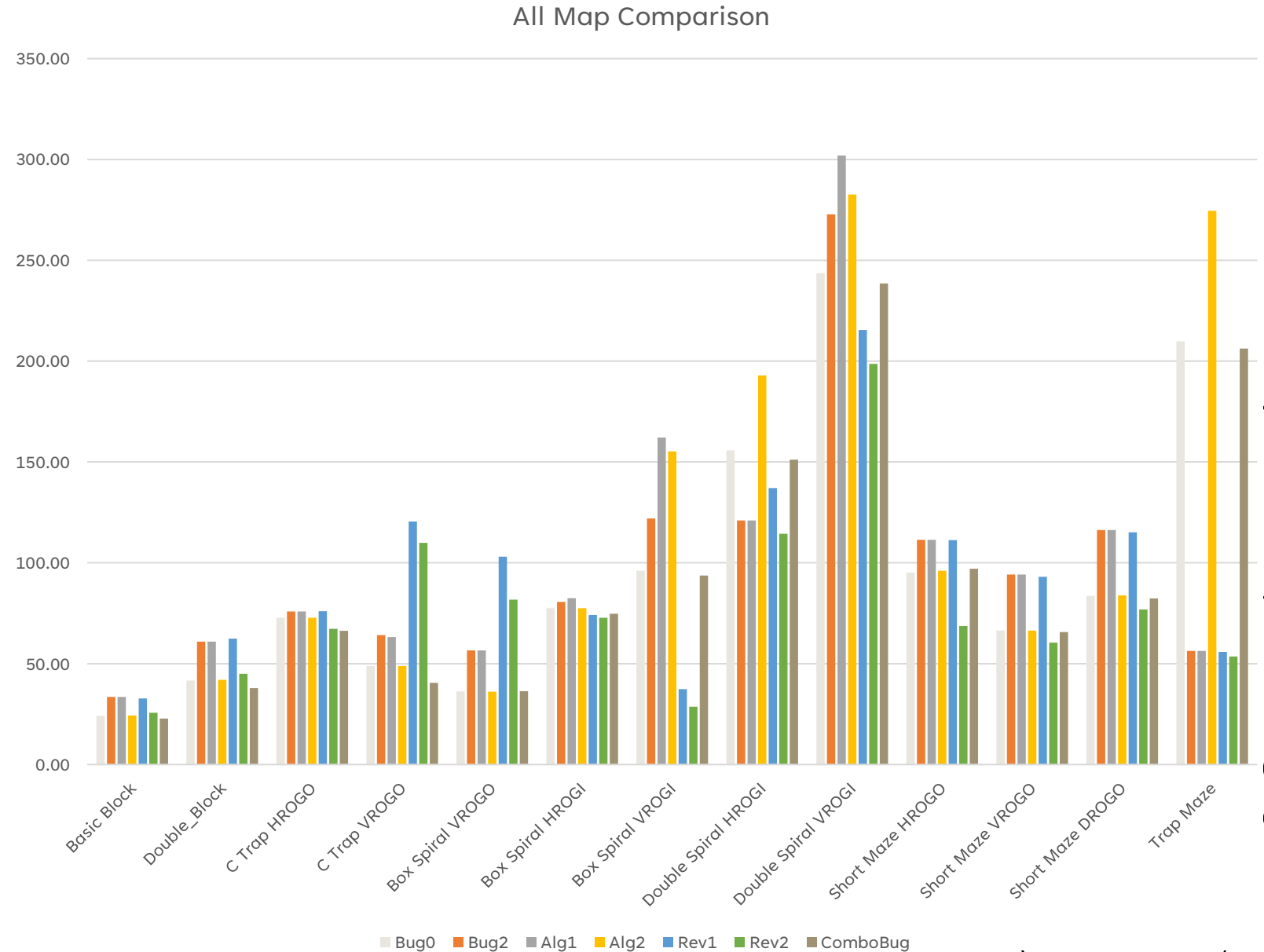
TESTING

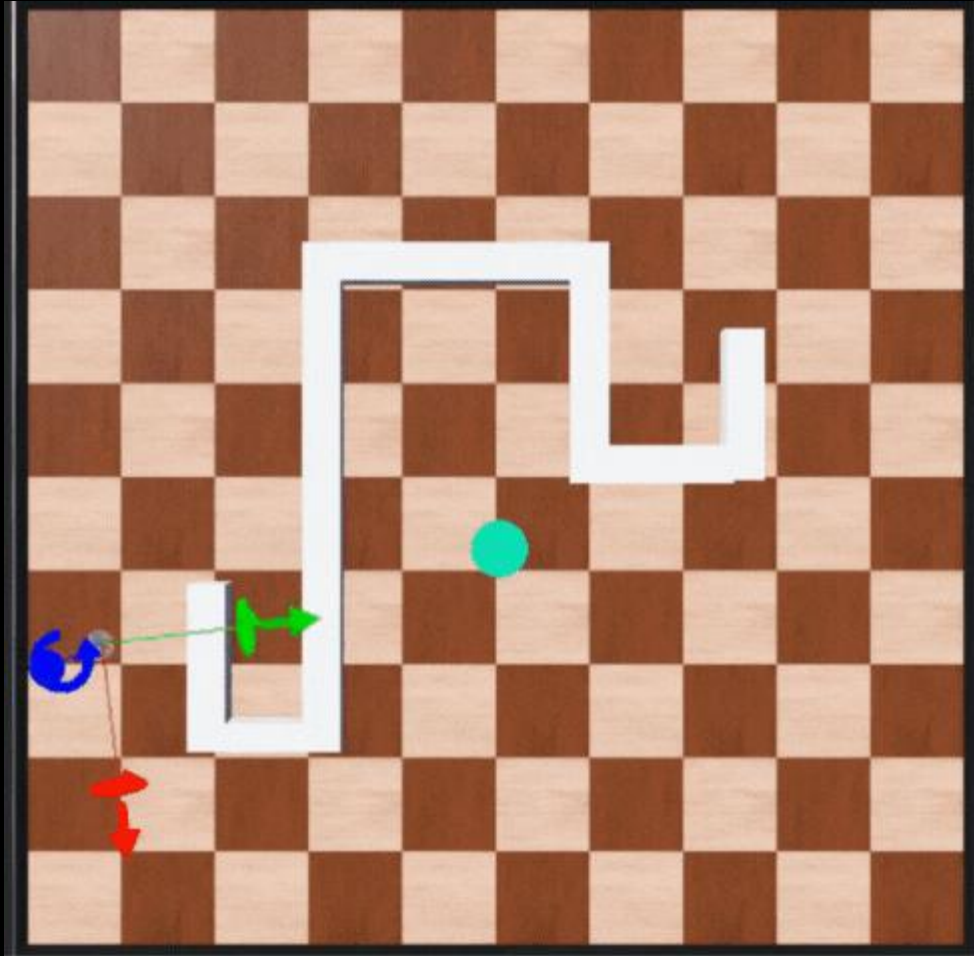
- Simulated in Webots
- Against common-sense and m-line algorithms
- Multiple obstacles ranging from simple to complex
- Recorded time to goal as a variable



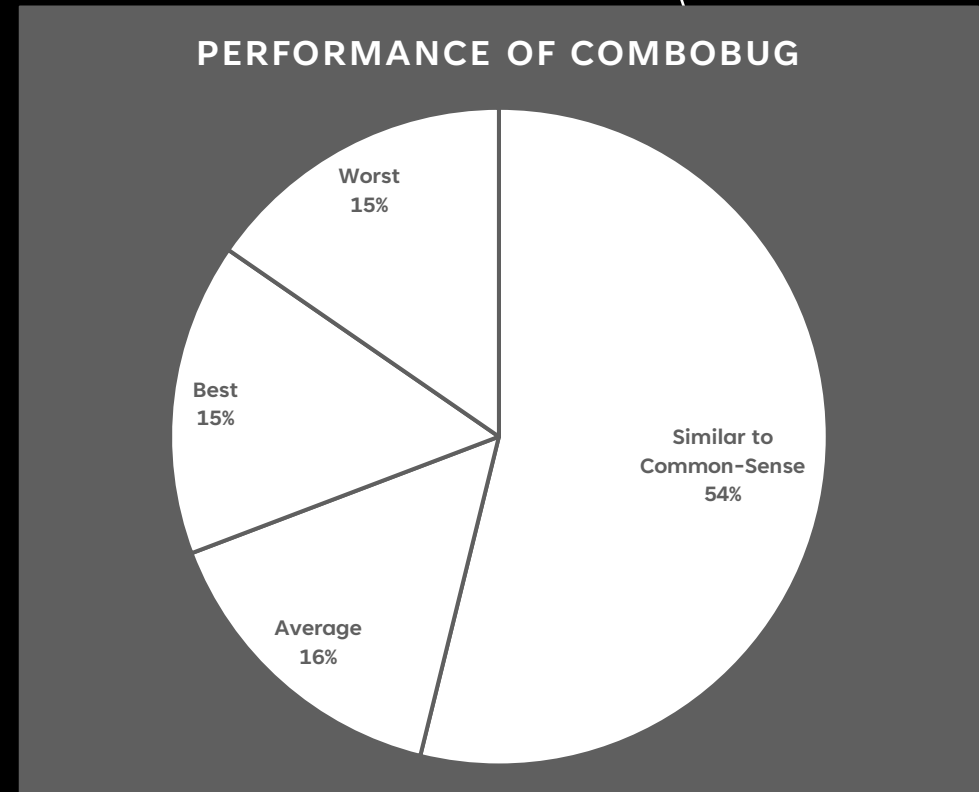
RESULTS

- The graph displays how often recorded time for ComboBug aligned with common-sense
- Use of m-line paths created and displayed new behavior but not better path time
- Recorded time is shorter in some cases, usually ones where common-sense succeeds.



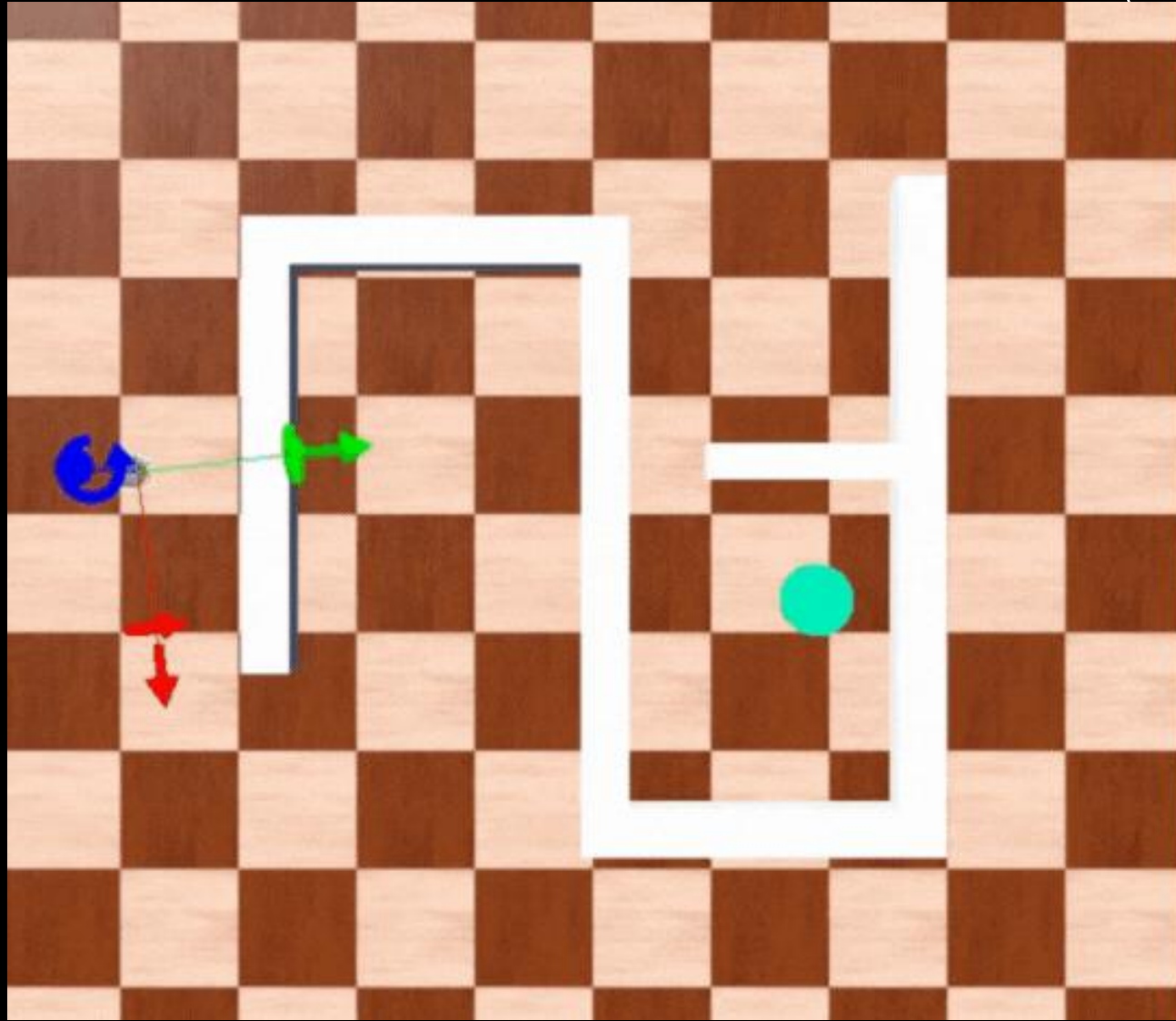


A display of ComboBug that appears as Bug0

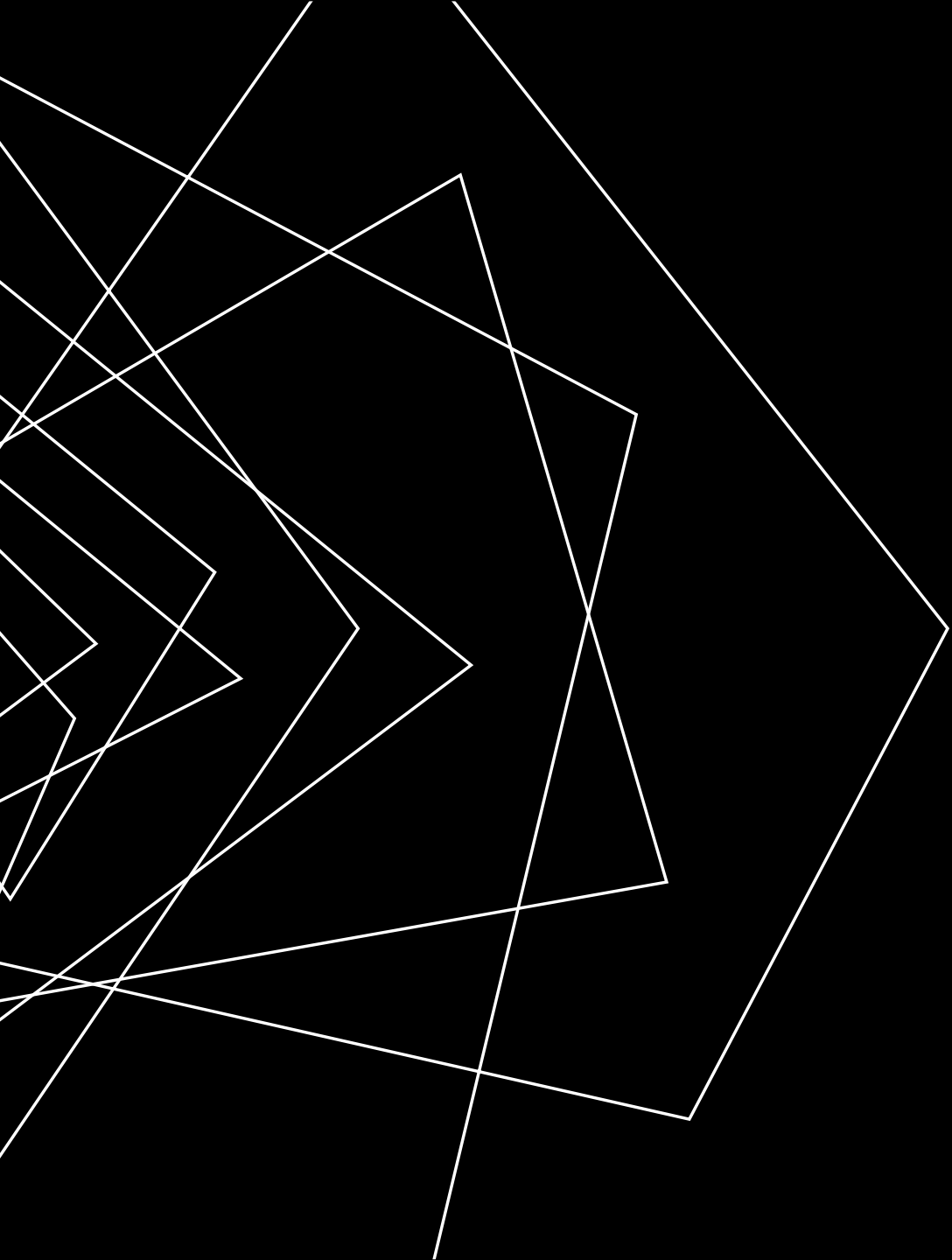


COMBOBUG AT WORK

A display of
ComboBug that uses
m-line.



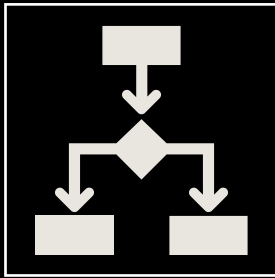
COMBOBUG AT WORK



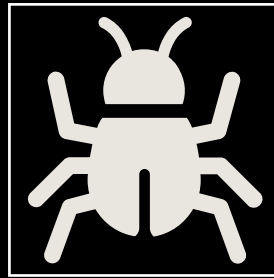
IN CONCLUSION

- The ComboBug successfully merges the M-Line and Common-Sense strategies
- Provides an algorithm that deterministically changes between strategies
- Is not an algorithm that produces shorter times but instead provides a 'back up' path.

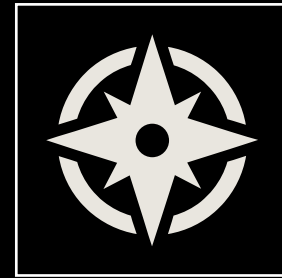
FUTURE WORKS



Move Towards
ComboBug Extensions



Bug Algorithm Based
AI Obstacle Avoidance



Real World Navigation
of ComboBug

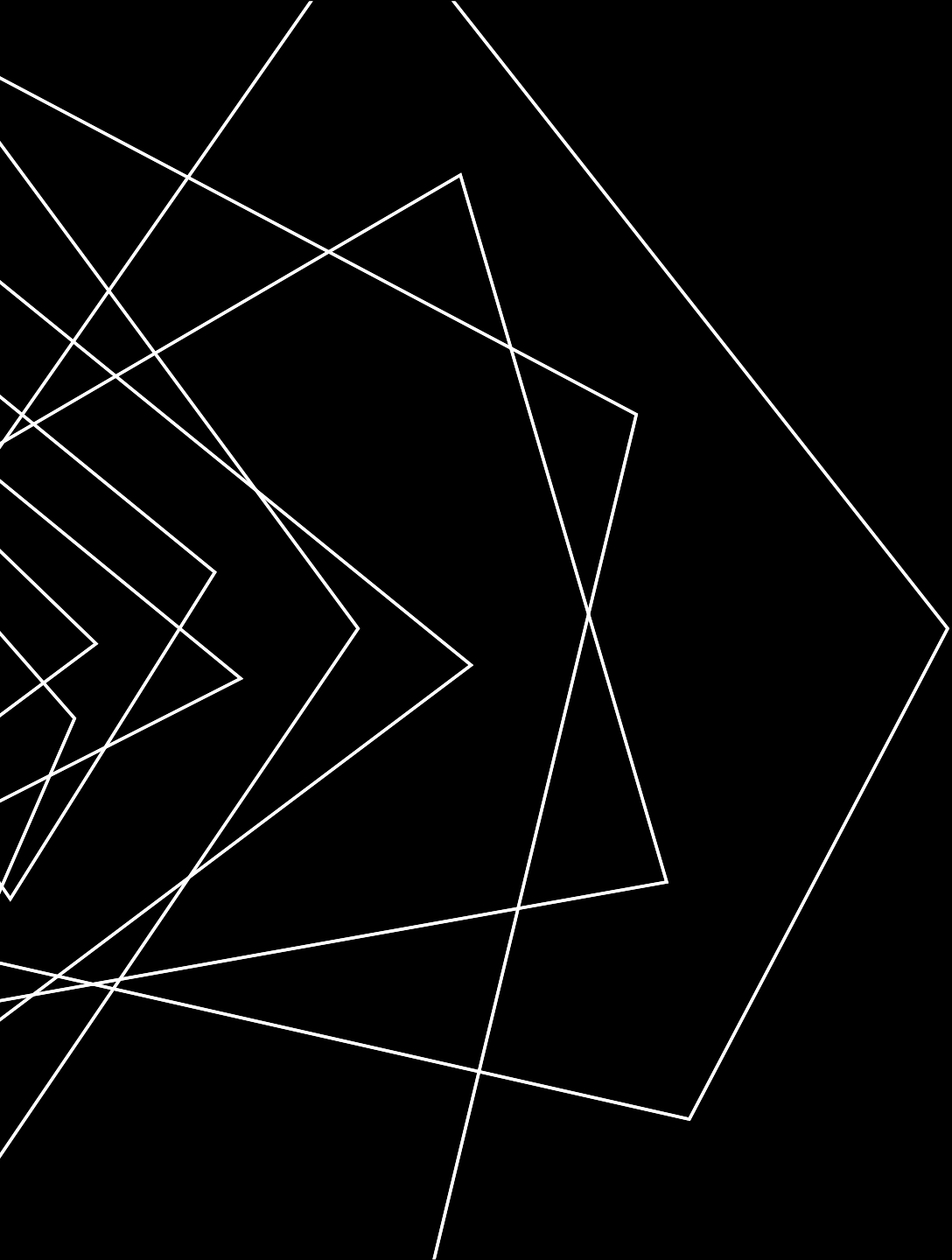
REFERENCES

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Yufka, Alpaslan, and Osman Parlaktuna. "Performance Comparison of the BUG's Algorithms for Mobile Robots." 2009.

J. Antich, A. Ortiz, and J. Mínguez, "A Bug-inspired algorithm for efficient anytime path planning," 2009 IEEE/RSJ International Conference on Intelligent Robots and Systems. 2009.

Y. Zhu, T. Zhang, J. Song, and X. Li, "A New Bug-type Navigation Algorithm Considering Practical Implementation Issues for Mobile Robots.

An abstract graphic on the left side of the slide, consisting of several white lines of varying lengths and orientations that intersect to form a series of overlapping, irregular polygons and triangles. The lines are thin and white, contrasting sharply with the black background.

THANK YOU

MARS Lab

Dr. Qi Lu

Izabella Valero

izabella.valero@utrgv.edu